Lecture 10: Linear transformations and matrices

$$T : \mathbb{R}^4 \longrightarrow \mathbb{R}^3$$

domain (inputs) — codomain (outputs)

$$T(cv + dw) = cT(v) + dT(w).$$

Standard (basis) vectors in $\mathbb{R}^4$:

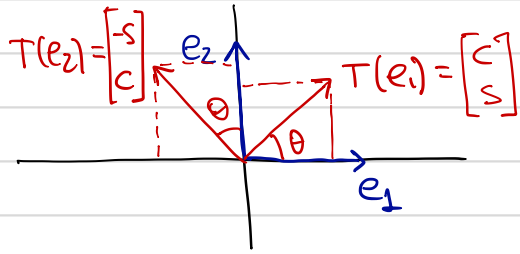
$$e_1 = \begin{bmatrix}1\\0\\0\\0\end{bmatrix}, \quad e_2 = \begin{bmatrix}0\\1\\0\\0\end{bmatrix}, \quad e_3 = \begin{bmatrix}0\\0\\1\\0\end{bmatrix}, \quad e_4 = \begin{bmatrix}0\\0\\0\\1\end{bmatrix}.$$

$$A = \begin{bmatrix} | & | & | & | \\ T(e_1) & T(e_2) & T(e_3) & T(e_4) \\ | & | & | & | \end{bmatrix}$$

$$v = \begin{bmatrix}1\\2\\3\\4\end{bmatrix} \Rightarrow Tv = T(e_1 + 2e_2 + 3e_3 + 3e_4)$$
$$= T(e_1) + 2T(e_2) + 3T(e_3) + 4T(e_4)$$
$$= Av.$$

Theorem: Every linear transf $T : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ comes from some $m \times n$ matrix A: $T(v) = Av$ for all v.

Example: $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ rotation by θ



$c = \cos(\theta)$
$s = \sin(\theta)$

$$A = \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \qquad T\left(\begin{bmatrix}1\\1\end{bmatrix}\right) = A \cdot \begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}c-s\\c+s\end{bmatrix}.$$

Terminology: $T : \mathbb{R}^n \longrightarrow \mathbb{R}^m$

- $T(v)$ is the image of v under T.
- range(T) = all vectors in $\mathbb{R}^m$ that are the images under T of some v in $\mathbb{R}^n$.

Example: $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$, $T(v) = A \cdot v$

$A = \begin{bmatrix} 1 & -2 & 4 \\ 3 & 0 & -5 \end{bmatrix}$. Is $w = \begin{bmatrix}3\\4\end{bmatrix}$ in range(T)? Yes!

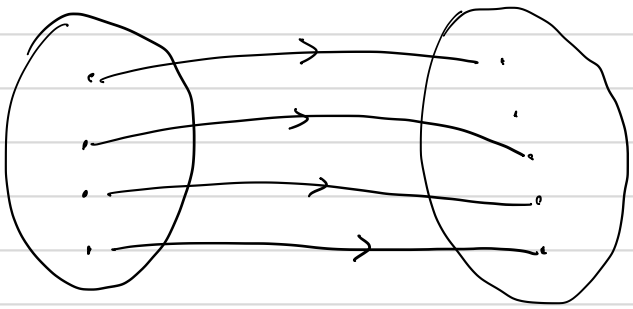
$$\left[\begin{array}{ccc|c} 1 & -2 & 4 & 3 \\ 3 & 0 & 5 & 4 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & -2 & 4 & 3 \\ 0 & +6 & -7 & -5 \end{array}\right]$$

$$\rightsquigarrow \quad v = \begin{bmatrix}4/3\\-5/6\\0\end{bmatrix} + s\begin{bmatrix}5/3\\7/6\\1\end{bmatrix}$$

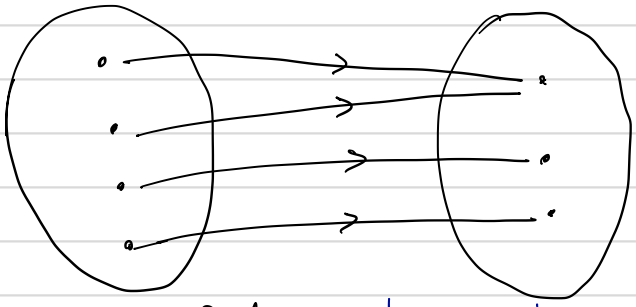
w is the image of ∞ many v's.

Def: $T: \mathbb{R}^n \to \mathbb{R}^m$ lin. transf.

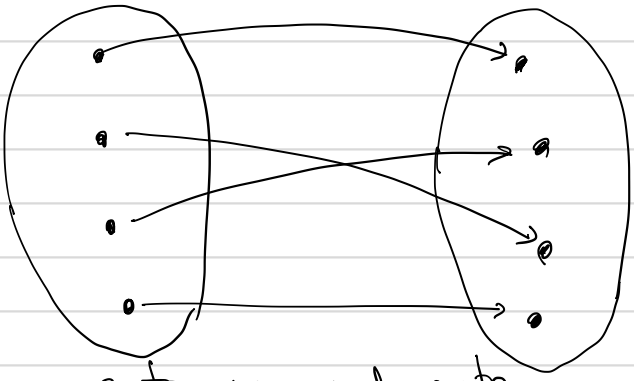
1) T is <u>one-to-one</u> if $T(u) = T(v)$ only when $u = v$.
2) T is <u>onto</u> if every $w \in \mathbb{R}^m$ is the image of some $u \in \mathbb{R}^n$ under T.



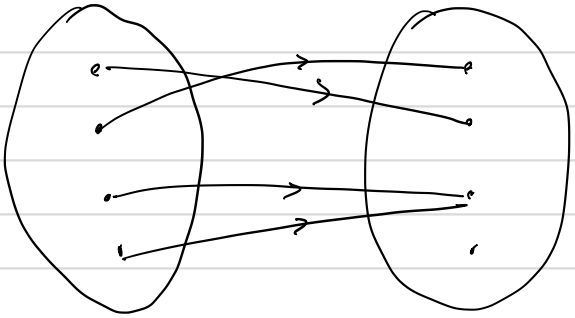
One-to-one, not onto



Onto, not one-to-one



one-to-one and onto



Not one-to-one, not onto.

Setup: A $m \times n$ matrix
$T: \mathbb{R}^n \to \mathbb{R}^m$ given by $T(v) = A \cdot v$.

Prop: TFAE:

1) cols of A span $\mathbb{R}^m$
2) $Ax = b$ has at least one sol. for all b.
3) T is onto.

Prop: TFAE:

1) cols of A are independent
2) $Ax = b$ has at most one sol. for all b.
3) T is one-to-one.

Obs:

1) If $m > n$, then T cannot be onto.
2) If $n > m$, then T cannot be one-to-one.

Thm: $(m = n)$ $T: \mathbb{R}^n \to \mathbb{R}^n$. TFAE:

1) cols of A span $\mathbb{R}^n$
2) cols of A are indep.
3) A is nonsingular
4) T is onto
5) T is one-to-one.